

Application of Uncertainty Propagation with Adaptive Gaussian Mixture Models for Cislunar Objects

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ABSTRACT

The focus of this paper is to develop an efficient and accurate method to propagate the Probability Density Function (PDF) representing the uncertain state of a spacecraft in cislunar space. Gaussian Mixture Models (GMMs) are a common method used to approximate an arbitrary PDF, with the added benefit of being convenient to use in an estimation filter. GMMs allow for more accurate uncertainty modeling in highly nonlinear scenarios with large observation gaps. Several studies have investigated adaptive GMMs, where a measure of nonlinearity is used to trigger splitting a component into several new components. In this paper we select two measures of nonlinearity based on statistical linearization error. An online merging mechanism is also developed based on the refinement of mixture entropy. The objective of the application of this adaptive GMM algorithm is the ease of implementation and tunability for a range of cislunar trajectories, such that a few intuitive split-merge parameters can be selected to perform uncertainty propagation.

1 INTRODUCTION

Recently there has been renewed interest in sending spacecraft to the Moon. Some notable recent cislunar missions include Artemis, CAPSTONE, Chandrayaan-1, Chandrayaan-2, Chang'e 5, Chang'e 6, and Lunar Flashlight. A new phenomenon is the interest from commercial enterprises in launching into the cislunar and lunar domain. The increasing number of spacecraft in this difficult dynamic domain is driving a need for more robust methods for maintaining Space Situational Awareness (SSA).

Cislunar SSA involves new challenges for space-object tracking; chief amongst them is uncertainty characterization. Typical practices for uncertainty propagation of Earth dominated orbits (e.g., covariance propagated with a state transition matrix) are inadequate for cislunar applications due to the highly nonlinear behavior caused by the gravitational interactions from the Earth, Sun, and Moon. Moreover, measurement gaps spanning from days to weeks amplify the need for accurate uncertainty propagation.

The nonlinear dynamics in cislunar space, coupled with infrequent observations, render unimodal Gaussian PDF representations of the uncertainty ineffective. Additionally, an uncertainty representation of an object that has a close approach with the Moon can quickly morph into a complicated and arbitrary shape. Several Bayesian GMM and particle filters provide a variety of parametrizations that offer particular advantages in the form of PDF characterization accuracy and/or speed of computation. One way to handle this situation is with a Gaussian mixture, where N Gaussian components can be used to represent any distribution with sufficient N . Adaptive GMM (AGM) based algorithms have been explored, of which a few examples are in [1, 2, 3, 4, 5, 6]. A few applications have employed a Gaussian mixture refinement with a weight update procedure by evaluating the time evolution of the mixture according to the Fokker-Planck-Kolmogorov Equation (FPKE) for continuous-time dynamical systems or the Chapman-Kolmogorov Equation (CKE) for discrete-time dynamical systems [7, 3]. An alternate approach to Gaussian mixture refinement is through splitting techniques, for example by detecting nonlinearity through monitoring of the deviation of the temporal derivative of the Differential Entropy [2]. Other examples of splitting methods have also been investigated involving monitoring the amount of L^2 error from the FPKE [3] or CKE [8]. The statistical linearization error described in [9] spawned several metrics [5, 10, 11]. In the context of splitting techniques, the decision of when to split and the direction of splitting directly influences the GMM refinement [12, 13, 14]. Several examples of splitting direction selection methods are available, where common practice is to choose a direction based on the splitting component's original second moment with largest eigenvalue [2] or a direction selected based on the maximum nonlinearity detected [13]. Refinement of the splitting process can be achieved through iterative methods as in [14]. Additional directional criteria are explored in [12].

A drawback to using a GMM is the expense of propagating a large number N of components. If using an Unscented Transform (UT) composed of n_{sp} sigma points, the representation of the GMM is composed of $(n_{sp} \times N)$ sigma points, exacerbating the propagation cost. Some forms of AGM algorithms also make use of a split-merge mechanism for GMM refinement [3] to enable tractability in the number of components. More complex merging algorithms such as the Gaussian Mixture Reduction by Clustering (GMRC) [15] or Constrained Optimized Weight Adaptation (COWA) [16] utilize compounding steps that lead to an optimized mixture [17]. In practice however, an AGM implementation requires tuning to obtain a tractable balance between computational load and uncertainty accuracy. The difficulty lies in determining this balance according to a splitting threshold and merging threshold before performing an analysis, which is primarily dependent on the size of initial uncertainty and the severity of nonlinearities present throughout the trajectory.

This paper develops an AGM process that utilizes a split-merge approach for scenarios in the cislunar domain. The relevant methodologies will be described in the following sections, followed by the results of testing on several cislunar scenarios simulated using high fidelity dynamics. A primary goal of this work is to incorporate the process into operational tracking applications, requiring efficient implementation. It is also highly desirable to have trajectory-agnostic settings that will work for any cislunar application, thus avoiding the need to manually tune a filter run for every individual spacecraft. The focus of this work is on the propagation/prediction of the uncertainty over longer time spans (e.g., up to two weeks), leaving the end-to-end filter results for later work.

2 ADAPTIVE GAUSSIAN MIXTURES BASED ON STATISTICAL LINEARIZATION

A key driver in deciding how to implement the GMM is the need to maintain a tractable number of Gaussian components. For the case where a large number of components is generated, the computational burden of propagating the GMM can become comparable to utilizing a Monte-Carlo method. This paper integrates several algorithms to form an adaptive GMM that is robust while maintaining acceptable computational costs. The proposed approach will reduce the GMM to a minimal representation during stable (or near linear) regions of the trajectory and trigger splitting of components when needed to compensate for highly nonlinear dynamic regions.

Methods to increase the GMM resolution by splitting typically utilize information based on estimates of nonlinearities or statistical information. One example of a nonlinearity measure discussed in [12, 18, 19] is the utilization of the Jacobian from the dynamics to evaluate departure from linearity [20]. A differential geometry method relies on the Hessian of the dynamics to assess the curvature of the nonlinear function at a given point [19, 21]. Metrics based on higher order Taylor series expansion (TSE) are developed in [22] and [23], where the latter identifies tensor eigenpairs to measure nonlinearity. Examples of methods that measure the statistical linearization error are outlined in [5, 10, 19], which aim to measure the deviation from statistical linear regression. Accessing higher order statistical information can also be performed by analyzing skewness [14, 24]. In general, the splitting criteria consider the cumulative buildup of nonlinearities, and splitting is performed when a threshold is reached. However, the trigger does not necessarily rely solely on the nonlinearity but is also dependent on the coupling between the nonlinearity and the existing uncertainty distribution. It is not always clear, for example, how suddenly a transition from Gaussian to non-Gaussian may happen, as each trajectory is different [25]. Therefore, threshold values for many such measures can often be only locally defined and difficult to interpret, potentially requiring a unique analysis for each orbit or transfer scenario.

A particularly attractive feature of some of the statistical linearization error metrics v such as the Mean Square Error (MSE) in [10, 11] or one shown by [5], both of which will be discussed in the following section, is their bounded nature $v \in [0,1]$. An additional advantage is that implementation of v is independent of the dynamics, so that low or high-fidelity models can be used interchangeably without modification. It is especially convenient when dealing with a variety of propagation tools and frameworks, where Jacobian or higher order TSE derived terms can require custom solutions. For trajectories traversing various dynamical environments, it can be useful to adjust the dynamics model to perform more accurate uncertainty propagation according to length of propagation and initial uncertainty [26, 27, 28].

When splitting is indicated by the trigger, the component experiencing the most nonlinearity is split in the direction of the maximum nonlinearity, according to a specific form of nonlinearity quantification [12, 13, 14]. Steps must be taken to limit the growth in the number of components; hence, a merging process is implemented to combine components that are similar per a Kullback-Leibler (KL) upper bound metric. The merging process leverages the local information given by analyzing any two candidate components, while also accounting for the overall information

given by the mixture. The merging process considered in this paper is designed to reduce the computational load while minimizing the loss of information.

2.1 Gaussian Mixture Representations

The PDF of a continuous random vector $\mathbf{x} \in \mathbb{R}^{n_d}$ with dimension n_d describes the relative likelihood of the random variable across regions in $\mathbb{R}^{n_d}$. A widely used PDF is the Gaussian PDF, which can be fully described by its mean $\mathbf{x} \in \mathbb{R}^{n_d}$ and covariance $\mathbf{P} \in \mathbb{R}^{n_d \times n_d}$. The PDF for $\mathbf{x}$ is defined as

$$p_g(\mathbf{x}; \mathbf{m}_x, \mathbf{P}_x) = \frac{1}{|2\pi\mathbf{P}_x|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \mathbf{m}_x)^T \mathbf{P}_x^{-1}(\mathbf{x} - \mathbf{m}_x)\right\} \quad (1)$$

An extension of the Gaussian PDF is the Gaussian mixture model (GMM) PDF, which is a weighted sum of Gaussian PDFs. The GMM is described by

$$\tilde{p}(\mathbf{x}) = \sum_{i=1}^L \alpha_i p_g(\mathbf{x}; \mathbf{m}_{x,i}, \mathbf{P}_{x,i}) \quad (2)$$

where L represents the number of components of the GMM, α_i represent the weights associated with each component, and $\mathbf{m}_{x,i}$ and $\mathbf{P}_{x,i}$ are the means and covariances for each component.

2.2 Statistical Linearization Error Criteria

The proposed adaptive GMM utilizes a splitting scheme based on statistical linearization regression method presented in [5]. Nonlinear state propagation is represented by the function $f(\cdot)$, evaluated via the UT. The nonlinear function can be approximated by the first-order Taylor series expansion around the first moment. The linearization error can then be utilized as a trigger mechanism, where larger nonlinearities will yield larger errors. From [5, 9], the statistical linearization is composed of two parameters, the matrix $\mathbf{F}$ and vector $\mathbf{b}$ such that,

$$\mathbf{y} = f(\mathbf{x}) \approx \mathbf{F}\mathbf{x} + \mathbf{b} \quad (3)$$

where the prior distribution of $\mathbf{x}$ is taken through the nonlinear transformation, resulting in the distribution $\mathbf{y} \in \mathbb{R}^{n_d}$. The approximation error term $\mathbf{e}$ is defined by

$$\mathbf{e} = f(\mathbf{x}) - (\mathbf{F}\mathbf{x} + \mathbf{b}) \quad (4)$$

This error describes the deviation between the nonlinear transformation and the linear approximation, taking into account the prior distribution of $\mathbf{x}$. The nonlinear transformation $f(\mathbf{x})$ can be evaluated with a UT [9, 29, 30] or quadrature-based approach [31, 32]. The parameters $\mathbf{F}$ and $\mathbf{b}$ are found via a minimization of the weighted sum of squared errors resulting in the following [5, 9],

$$\mathbf{F} = \mathbf{P}_{xy}^T \mathbf{P}_x^{-1}, \quad \mathbf{b} = \mathbf{m}_y - \mathbf{F} \cdot \mathbf{m}_x \quad (5)$$

where $\mathbf{m}_x$ and $\mathbf{P}_x$ denote the mean and covariance of $\mathbf{x}$, $\mathbf{P}_{xy}$ denotes the cross-covariance matrix of $\mathbf{x}$ and $\mathbf{y}$, and $\mathbf{m}_y$ is the mean of $\mathbf{y}$. The linearization error is characterized by the error term $\mathbf{e}$, which has zero mean and a covariance matrix given by

$$E[\mathbf{e}\mathbf{e}^T] = \mathbf{P}_e = \mathbf{P}_y - \mathbf{F}\mathbf{P}_x\mathbf{F}^T \quad (6)$$

where $\mathbf{P}_y$ is the covariance of $\mathbf{y}$, computed via the UT. In the application of the linearization error, we look at two different splitting criteria, v_{MSE} and v_{EXP} , both of which depend on the trace of the error covariance:

$$\epsilon = \text{tr}(\mathbf{P}_e) \in [0, \infty) \quad (7)$$

The first form of normalization is demonstrated in [10, 18, 19] and referred to as the mean square error (MSE). The normalization utilizes the trace of the covariance $\mathbf{P}_y$ and provides a bounded response.

$$v_{MSE} = \sqrt{\frac{\epsilon}{\text{tr}(\mathbf{P}_y)}} \in [0,1] \quad (8)$$

The second measure, v_{EXP} , is presented by Huber [5], where the exponent of the errors is considered along with the weight of the component under consideration. The exponential leads to a more sensitive response with a tuning parameter to bias splitting toward higher weighted components. The metric utilizes an exponential functional to construct the splitting criterion v_{EXP} .

$$v_{EXP} = \alpha^\gamma \cdot (1 - \exp(-\epsilon))^{1-\gamma} \in [0,1] \quad (9)$$

The parameter $\gamma \in [0,1]$ determines the relative importance of the component weight. In this paper, the component weight α is not considered for the splitting criterion v_{EXP} ; therefore, γ is set to zero.

2.3 Splitting Components

Once nonlinear error growth in a Gaussian component indicates splitting, a splitting direction and number of new components must be chosen. Typically, there are strong nonlinearities in a particular direction, contraindicating a split in all directions. A simple approach is to split in the direction of the maximum covariance eigenvector, which assumes that the covariance is growing fastest in the direction of nonlinearities. There are more sophisticated methods, such as checking specific directions to see which causes the largest deviation in states due to nonlinear forces acting upon them [5, 13]. This work uses the splitting direction found via the method described in [13]. The j^{th} potential splitting direction is ranked based on a measure of sensitivity of the PDF to the underlying dynamic propagation:

$$NL_{i,j}(\hat{\mathbf{a}}_j) = \frac{f(\mathbf{m}_i + \tilde{h}\sigma_{\hat{\mathbf{a}}_j}\hat{\mathbf{a}}_j) + f(\mathbf{m}_i - \tilde{h}\sigma_{\hat{\mathbf{a}}_j}\hat{\mathbf{a}}_j) - 2f(\mathbf{m}_i)}{2\tilde{h}^2} \quad (10)$$

$$\sigma_{\hat{\mathbf{a}}_j} = \|(\mathbf{S}_i)^{-1}\hat{\mathbf{a}}\|_2^{-1} \quad (11)$$

where $\hat{\mathbf{a}}_j$ is a unit vector for an arbitrary split direction, $f(\cdot)$ is the propagation function, $\mathbf{S}_i$ is the square-root factor of the covariance matrix $\mathbf{P}_i$, and $\tilde{h} = \sqrt{3}$ is recommended for Gaussian distributions [13].

The eigenvectors of the covariance matrix $\mathbf{P}$ are calculated and each direction is tested via Eq. (10). The direction with the maximum p-norm measure of NL_j is chosen as the splitting direction, where in this work the Euclidean norm $|NL_j|$ is chosen. Jones [26] utilizes Eq. (10) and shows that the maximum eigenvector corresponded to the $\max(|NL|)$ direction. Since that is not guaranteed to be true for continued propagation, a forward propagation on each direction is performed to determine the direction of split. As an aside for multi-directional splitting, the descending order of the covariance eigenvectors does not always match the ranking according to NL [26].

A splitting library is used to distribute N_s gaussian components that will match the gaussian PDF of the original component when summed [2, 13]. This effort utilizes the 3-component splitting library from [13], that is archived online¹. Most splitting libraries identified, including [13], assume homoscedastic splitting, where each new component has the same variance.

2.4 Adaptive Merging

A robust merging process is needed to maintain performance and prevent an overgrowth in the number of GMM components. Merging can be a tricky process; as without appropriate settings, a scenario may arise where the merging process undoes a splitting operation. This is obviously undesirable, as the splitting was indicated due to the presence of nonlinear forces nullifying the assumptions placed on uncertainty propagation. In addition, each merging operation will cause some degree of information loss due to the reduction in the number of components. Another issue concerns

¹ <http://russell.ae.utexas.edu/code/GMMsplittingLibrary.txt/>

the generation of sigma points representing the new component; by eliminating the original (pre-merge) sigma points, any accumulated nonlinearities will be discarded.

The process adopted in this work includes multiple steps to minimize information loss. An adaptive merging threshold value is used to initiate merging which is driven by the amount of overlapping information present in the GMM. A test is performed to assess the minimum number of components to reach in the merging process without losing excessive information. Further, a step is taken to prevent the recombination of recently split components even if the minimum merging criteria indicates additional merging. Merging is then performed based on Runnalls' algorithm, where a measure of similarity between two components is based on the KL discrimination measure [33]. There are more sophisticated methods available to perform merging [16, 34, 35, 36], e.g. Gaussian Mixture Reduction via Clustering (GMRC) [15]. However, in testing it was found that the optimization steps of the GRMC method do not provide significant benefits relative to the added cost of the optimization. Simply using the Runnalls method gives valuable results with minimal computation cost relative to the GMRC method, though the reduced mixture may be represented by a different set of components.

2.4.1 Merging Algorithm

The objective of a merging procedure is to minimize the dissimilarity between the original (pre-merged) and merged mixtures. Among methods of merging in which a single criterion is used [33, 37, 38], Runnalls [33] proposes a criterion for merging based on the KL discrimination. In terms of dissimilarity using KL between any two mixtures, a closed-form expression is not available. Instead, Runnalls defines a pairwise upper-bound KL metric between a post-merged Gaussian and two pre-merged components. The upper-bound KL is

$$D_{uKL}(p_{g,i}, p_{g,j}) = \frac{1}{2} \left((\alpha_i + \alpha_j) \log(|\mathbf{P}_{x,ij}|) - \alpha_i \log(|\mathbf{P}_{x,i}|) - \alpha_j \log(|\mathbf{P}_{x,j}|) \right) \quad (12)$$

where $p_{g,i}$ and $p_{g,j}$ are components i and j with weights α_i and α_j . The covariance $\mathbf{P}_{x,ij}$ is the result of merging components i and j [33]. Merging two components results in:

$$\alpha_{ij} = \alpha_i + \alpha_j \quad (13)$$

$$m_{ij} = \frac{\alpha_i}{\alpha_{ij}} m_i + \frac{\alpha_j}{\alpha_{ij}} m_j \quad (14)$$

$$P_{ij} = \frac{\alpha_i}{\alpha_{ij}} P_i + \frac{\alpha_j}{\alpha_{ij}} P_j + \frac{\alpha_i \alpha_j}{(\alpha_{ij})^2} (m_i - m_j)(m_i - m_j)^T \quad (15)$$

A property of D_{uKL} is that, unlike the formal KL divergence, the function is symmetric, such that $D_{uKL}(i, j) = D_{uKL}(j, i)$. This property is useful, because only the upper triangular of the pair-wise combination matrix is needed, allowing for increased computational efficiency. The D_{uKL} threshold describes the cost of merging components i and j , with a merge proceeding when D_{uKL} is under some threshold value; a low threshold value is more conservative and keeps the pre-merge mixture intact, while a high threshold value allows for increased merging. The implementation of Runnalls in this work indicates merging until no component pairs evaluated are below the threshold value.

2.4.2 Mixture Size Optimization

To maintain tractability of the GMM PDF during propagation, the mixture is evaluated at each time step to determine its underlying structure. The method is inspired by a merging approach devised by [3], which considers the similarity in information between any two components and the relative orientation. However, here the approach is to leverage the structure of the mixture. The structure of the mixture is analyzed with an approximation procedure using differential entropy for a Gaussian mixture [39] to determine an upper bound on the merging trigger resulting in a minimum mixture size while not losing excessive information.

The entropy generally cannot be calculated in closed form for Gaussian mixtures. Therefore, a scheme for entropy approximation is required. Huber et al. [39] describes a computationally effective approach to compute the mixture entropy upper bound. Starting with the definition of entropy,

$$H(\mathbf{x}) = - \int_{\mathcal{S}} p(\mathbf{x}) \log p(\mathbf{x}) d\mathbf{x} \quad (16)$$

The mixture entropy is defined by substituting the Gaussian mixture distribution $\tilde{p}(\mathbf{x})$ for $p(\mathbf{x})$ in Eq. 16:

$$H(\mathbf{x}) = - \int \sum_{i=1}^L \alpha_i \cdot p_g(\mathbf{x}; \mathbf{m}_{x,i}, \mathbf{P}_{x,i}) \cdot \log \left(\sum_{j=1}^L \alpha_j \cdot p_g(\mathbf{x}; \mathbf{m}_{x,j}, \mathbf{P}_{x,j}) \right) d\mathbf{x} \quad (17)$$

The upper bound $H_u(\mathbf{x})$ is demonstrated by Huber et al. [39] as

$$H_u(\mathbf{x}) = \sum_{i=1}^L \alpha_i \cdot \left(-\log \alpha_i + \frac{1}{2} \log((2\pi e)^{n_d} |\mathbf{P}_{x,i}|) \right) \quad (18)$$

The structure of the mixture can be revealed by an upper bound refinement. For the case of a large number of components, the shape of the mixture can be reduced to a subset of components that sufficiently captures the various modes presented by $\tilde{p}(\mathbf{x})$. The procedure to perform upper bound refinement is conducted through a series of merging steps similar to Algorithm 1 in [39]. The refinement procedure is condensed to a process that finds the minimum $H_u(\mathbf{x})$ through an optimization

$$\mathcal{D}^* = \arg \min_{\mathcal{D}} H_u(\mathbf{x}) \quad (19)$$

More precisely, an iterative optimization controlled by a variable $\mathcal{D}$, i.e., the Runnalls ($\mathcal{D} = D_{uKL}$) merge trigger from Eq. (12), is employed to determine the lower bound $H_u^*(\mathbf{x})$. In the initialization step, $\mathcal{D}_0$ represents a value that incurs no merging, and $H_u(\mathbf{x})$ is computed for the given mixture. For each incremental merging step, such that $\Delta\mathcal{D} + \mathcal{D}$, the value $H_u(\mathbf{x})$ initially decreases until a minimum is found, denoted $H_u^*(\mathbf{x})$. For additional merging steps past $H_u^*(\mathbf{x})$, the increase in $H_u(\mathbf{x})$ suggests that too few components are likely to represent the GMM. Notably, the optimization process described can utilize any merging algorithm.

The optimization of Eq. (19) uses the merging procedure discussed in Section 2.4.1 to determine $\mathcal{D}^*$. However, it is not always recommended to reach H_u^* , because it is possible to discard too much accumulated uncertainty information. (This is not a consideration during the optimization.) Therefore, a preservation variable $\Gamma_p \in [0,1]$ is applied after optimization to control the amount of merging.

$$\mathcal{D}_s = (1 - \Gamma_p) \mathcal{D}^* \quad (20)$$

For example, in the case $\Gamma_p = 1$, no merging occurs. Conversely, in the case $\Gamma_p = 0$, merging occurs maximally according to H_u^* with $\mathcal{D}^*$.

To illustrate the mixture size optimization with $\mathcal{D}^*$ and $\mathcal{D}_s$, the method is applied to the High-Earth Orbit example presented in [2] with the same initial single input Gaussian, using v_{MSE} splitting approach from Eq. (8). The uncertainty GMM is propagated with the splitting threshold $v_{MSE} = 0.011$, with no merging. The mixture size optimization is performed at the final time of $t_f = 1.51$ days. Fig. 1 presents a comparison of the mixture size optimization achieved with the D_{uKL}^* and that achieved by D_{uKL} with $\Gamma_p = 0.9$. The GMM grows to 63 components via splitting at t_f . The reduced GMM with D_{uKL}^* is composed of three components. The reduced GMM with Γ_p applied is composed of eight components. It is seen that the reduced GMM with D_{uKL}^* does not capture the edges well compared to the original, while the use of Γ_p results in a GMM that captures the majority of the detail on the edges with significantly fewer components than the original mixture.

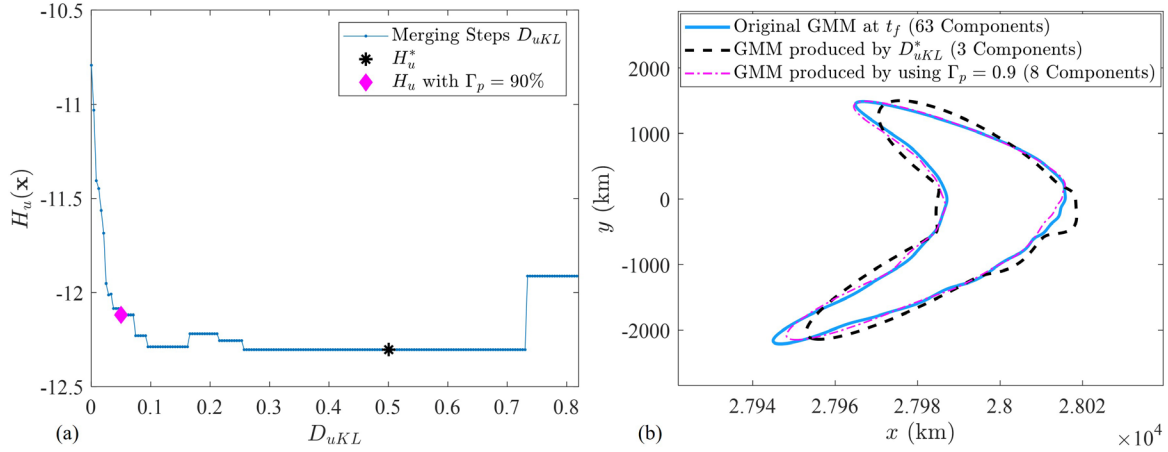


Fig. 1: (a) Plot of the $H_u(\mathbf{x})$ as a function of the merging threshold D_{uKL} . $H_u^*(\mathbf{x})$ is denoted by the black asterisk and the $H_u(\mathbf{x})$ resulting from $\Gamma_p = 0.9$ is the magenta diamond. (b) The propagated GMM projected onto the x - y plane showing the 99.9% probability contour. The full GMM without merging is shown in blue, the GMM as a result of D_{uKL}^* is shown by the dashed black contour, and the GMM with $\Gamma_p = 0.9$ is shown by the magenta dot-dashed line.

2.4.3 Component Exclusion

For any time step in which a split is performed on any component, it is important to account for the split information in the merging process to avoid nullifying a split of the current time step. Therefore, prior to any merging, a lower bound threshold $\mathcal{D}_e$ is found based on any splits that occurred within the same time step. For the n th component that is split and generates sub-mixture $\tilde{p}_{\xi_n}(\mathbf{x})$ composed of M components,

$$\tilde{p}_{\xi_n}(\mathbf{x}) = \alpha_n p_g^{(n)}(\mathbf{x}) = \sum_{h=1}^M \alpha_h p_g(\mathbf{x}; \mathbf{m}_{x,h}, \mathbf{P}_{x,h}) \quad (21)$$

where the sub-mixture that replaced the weighted component is analyzed with the merging cost matrix $\mathcal{D}_{\xi_n}(i, j)$ for components i and j of each mixture. The cost of merging is computed for each i to j component in the new sub-mixture. Such that for n splits, $\Xi = \{\xi_1, \dots, \xi_n\}$ the merging cost is computed to determine a lower bound threshold in terms $\mathcal{D}$. The lower bound is found by taking the minimum cost over all merging costs for each sub-mixture $\tilde{p}_{\xi_n}(\mathbf{x})$. Fig. 2 provides an illustration of the process to determine the lower-bound threshold $\mathcal{D}_e$ based on n splits.

$$\mathcal{D}_e = \min \left\{ \mathcal{D}_{\xi_1}, \mathcal{D}_{\xi_2}, \dots, \mathcal{D}_{\xi_n} \right\} \quad (22)$$

The proposed exclusion method is not meant to restrict merging between any pre-split components in $\tilde{p}(\mathbf{x})$ with newly split components in $[\tilde{p}_{\xi_1}, \dots, \tilde{p}_{\xi_n}]$. The goal of the method is to restrict a merging of components within each new sub-mixture that results from a splitting operation. The merging threshold $\mathcal{D}_s$ is replaced by taking the minimum over $\mathcal{D}_s$ from Eq. (20) and $\mathcal{D}_e$ from Eq. (22) to avoid both over-merging in terms of the overall mixture and re-merging recently split components:

$$\mathcal{D}_s = \min[\mathcal{D}_s, \mathcal{D}_e] \quad (23)$$

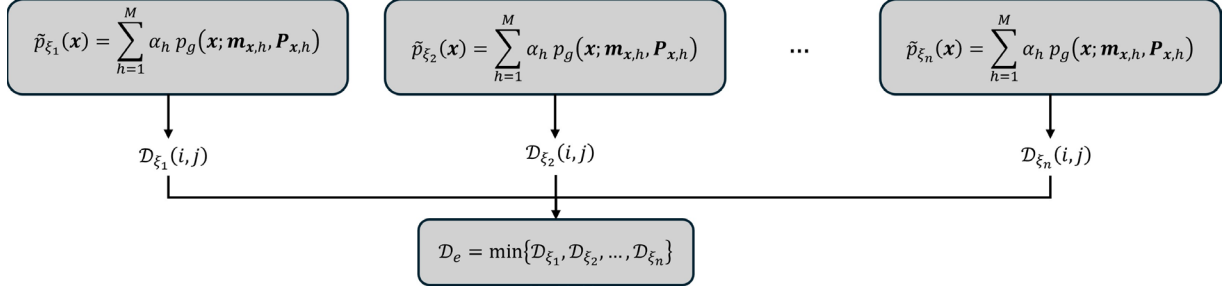


Fig. 2. Split-based component exclusion based on a set of splits $\Xi = \{\xi_1, \dots, \xi_n\}$. $\mathcal{D}_{\xi_n}(i, j)$ denotes the merging cost matrix for each n sub-mixture as a result of a split.

2.5 Monte Carlo and Performance Metrics

In order to compare the accuracy of the adaptive split and merge method proposed, the Monte Carlo method from [25] is utilized. Typically, for a state $n_d \geq 6$, Monte Carlo methods rely on a large number of samples; however, the computational load for each particle can prove expensive. Therefore, initialization of K particles is populated via the multivariate Gaussian particle cloud method from [25] that performs an optimization over a set of generated samples. The optimization constraints are based on the Cramér-von Mises metric, Mahalanobis distance, and skewness. For a set of K particles, the multivariate Gaussian particle cloud avoids the possible presence of gaps and clumps that make up an uneven distribution from using normally distributed random samples.

A comparison of a propagated GMM PDF to the set of propagated particles is performed in two ways. The first employs the commonly used Likelihood Agreement Measure (LAM) [2], which compares each component to each generated particle and determines the amount of overlap between the two PDFs $p(\mathbf{x})$ and $q(\mathbf{x})$. A LAM between two PDFs is:

$$l(p, q) = \int p(\mathbf{x})q(\mathbf{x}) d\mathbf{x} \quad (24)$$

The particle PDF denoted by $\tilde{q}(\mathbf{x})$ is given by a Dirac mixture model (DMM),

$$\tilde{q}(\mathbf{x}) = \sum_{i=1}^K \varphi_i \delta(\mathbf{x} - \boldsymbol{\mu}_{x,i}) \quad (25)$$

where the term $\delta(\mathbf{x} - \mathbf{x}_{p,i})$ is the Dirac delta distribution centered at each particle's state $\mathbf{x}_{p,i}$ with uniform weights $\varphi_i = 1/K$. When the GMM PDF represented by $\tilde{p}(\mathbf{x})$ is substituted for $p(\mathbf{x})$, and the DMM PDF $\tilde{q}(\mathbf{x})$ for $q(\mathbf{x})$, the LAM between the GMM and DMM PDFs is

$$l(p, q) = \sum_{i=1}^K \sum_{j=1}^L \varphi_i \alpha_j p_g(\mathbf{x}_{p,i}; \mathbf{m}_{x,j}, \mathbf{P}_{x,j}) \quad (26)$$

The second metric used is the Number Not Associated (NNA) provided in [40], which is a method based on data association. The data association is performed between the predicted GMM and measurements derived from particles without random noise. In this application, the number of particles that would not be associated in a GMM-based filter are counted. The measurement model used in this analysis is topocentric right-ascension and declination from an Earth-based observer. To evaluate the association of each particle to the GMM, a minimum Mahalanobis distance is computed across all components. For a given threshold on the Mahalanobis distance, the particle would either be rejected or accepted based on its minimum Mahalanobis distance. While the threshold can influence the result, the idea is to look for areas of the predicted GMM PDF that are too diffuse to make an effective update in the presence of a measurement.

A third metric is used when a merging analysis is performed to quantify the amount of information lost for each merging procedure from Section 2.4. The Normalized Integral Square Error (NISE) is used to compare between a pre-merge and post-merge mixture for each time step in which merging occurred. The NISE is defined by

$$J_{NISE} = \frac{\int (\tilde{p}_A(\mathbf{x}) - \tilde{p}_B(\mathbf{x}))^2 d\mathbf{x}}{\int (\tilde{p}_A(\mathbf{x}))^2 d\mathbf{x} + \int (\tilde{p}_B(\mathbf{x}))^2 d\mathbf{x}} \quad (27)$$

where the pre-merge mixture is defined by $\tilde{p}_A(\mathbf{x})$ and post-merge mixture $\tilde{p}_B(\mathbf{x})$.

2.5.1 Initial Uncertainty

The coupling of uncertainty growth and large propagation times for trajectories traversing a wide variety of nonlinearity strengths exacerbates the uncertainty quantification problem. To determine an initial covariance that is closer to operational cases, this work utilizes a least-squares orbit determination process similarly used by [40], where constrained angles-only measurements are generated based on the nominal state of the object. An Earth-based observer is selected based upon elevation and line-of-sight constraints, such that the solution is reflective of the majority of the trajectory [41]. The result of the least-squares orbit determination is representative of each trajectory. An appropriate scaling is applied for each trajectory to stress the split/merge algorithm appropriately, leveraging previous studies [6, 42].

3 TEST SCENARIOS

We study the effectiveness of the proposed method on three cislunar scenarios: a Distant Retrograde Orbit (DRO), a Lunar Transfer Orbit (LTO), and a Near Rectilinear Halo Orbit (NRHO). These orbits were chosen because they are characteristic of trajectories that have been utilized by operational satellites and because they present challenges to tracking and surveillance scenarios. The DRO is a relatively dynamically benign scenario and can be considered as a baseline. The LTO is difficult to track/predict because it crosses three dynamic regions: Earth departure where Earth gravity dominates, cislunar transfer where both Earth and Moon are important, and termination near the Moon where Lunar gravity dominates. The NRHO is difficult because it has relatively close approaches to the Moon which disrupt the uncertainty distribution, causing it to stretch out and nullify linear assumptions.

3.1 Trajectories and Adaptive Gaussian Mixture Parameters

To assess the performance of the adaptive Gaussian mixture algorithm presented in Section 2, three types of trajectories are tested. The reference orbits for the orbital trajectories are derived from the Earth-Moon Circular Restricted Three-Body Problem (CRTBP) and differentially corrected in the full-ephemeris dynamics model to obtain pseudo-periodic motion. The first trajectory is derived from a DRO in a nearly 2:1 sidereal resonance with the Moon, possessing a Jacobi constant of $J_C = 2.93$ and period of 13.62 days. The second trajectory is an LTO with duration of 5 days, an example of which can be found in [43]. The third trajectory is an NRHO in a nearly 3:1 synodic resonance with Jacobi constant $J_C = 2.998$ and period of 9.851 days. Both the DRO and NRHO trajectories were differentially corrected from the ideal conditions to produce suitable initial conditions to be used with a high-fidelity model.

The Earth and Moon states are defined according to the JPL planetary ephemerides DE440 [44]. The model incorporates an Earth gravity model EGM2008 4×4, Moon and Sun as point masses, and solar radiation pressure. The spacecraft model used for all trajectories utilizes an albedo-area-to-mass ratio such that $C_r A/m = 0.0109 \text{ m}^2/\text{kg}$. It is noted that varying high-fidelity models were explored but are not expanded upon in this study. To perform the PDF prediction, this paper utilizes the Turboprop C++ astrodynamics library accessed via a MATLAB API interface [45].

The three trajectories are displayed in Fig. 3 in the Earth-Moon rotating frame adjusted for a common epoch. The DRO and NRHO trajectories are propagated to a few periods. However, this analysis considered a 13-day propagation for the DRO, a 19-day propagation for the NRHO, and a 5-day propagation for the LTO.

A comparison is performed for the adaptive GM between the two statistical-linearization-based criteria v_{MSE} and v_{EXP} from Section 2.2. For all three trajectories, the split criteria were set to $v_{MSE} = 1.0 \times 10^{-3}$ and $v_{EXP} = 5.0 \times 10^{-2}$. The splitting direction was chosen based on Eq. (10) with a forward propagation defined by $t_s + 3\Delta t$ for each eigen direction, where t_s is the indicated component's split time and Δt is the propagation time step interval used to evaluate the splitting and merging criteria. The merging preservation variable was set to $\Gamma_s = 0.9$. The time step used to evaluate

the splitting and merging for the DRO and NRHO trajectories was $\Delta t = 30$ min, and for the LTO, $\Delta t = 6$ min. For initial uncertainty, a single input Gaussian from the least-squares orbit determination process described in Section 2.5.1, is used to initialize the simulation for each trajectory.

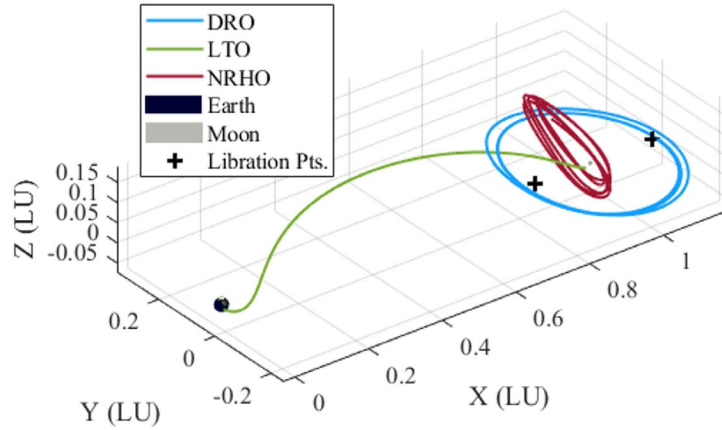


Fig. 3. Illustration of the DRO in blue, LTO in green and NRHO in red with Earth, Moon and the L_1 and L_2 libration points.

4 SIMULATION RESULTS

The adaptive split-merge algorithm was evaluated using the DRO, LTO and NRHO trajectories described above. The following sections examine each trajectory and the predicted GMM resulting from the v_{MSE} or v_{EXP} splitting criteria, utilizing the same merging criteria defined by Γ_p . For each case, a set of 10,000 Monte-Carlo particles is propagated to represent the true distribution for comparison purposes. For the performance metric NNA, at each trajectory's respective final time t_f , an Earth-based observer was selected to determine which particles did not associate. The 0.999 quantile for a chi-square distribution was used to determine if each particle was associated. Note that for this work, process noise was not considered.

4.1 Distant Retrograde Orbit Results

The results of the predicted GMM PDF at t_f using the statistical-linearization-based split criteria and the merging procedure are presented in Fig. 4 for the DRO. The main diagonal of subplots displays the PDF for each dimension $x \in \mathbb{R}^d$. The Monte Carlo particles are plotted in each subplot to show the true distribution. The lower triangular pair-wise plots display the projections of the overall GMM into the two coordinate directions indicated. The upper right trajectory plot depicts the blue DRO orbit trajectory considered for this analysis, the propagation end time t_f location on the DRO orbit is shown by the red point. At the end of the prediction, the GMMs encapsulate the majority of particles as seen in each projection.

The time-history of the number of components is shown in Fig. 5(a), where the spikes indicate that a merging occurs immediately after a splitting event. Focusing on v_{MSE} GMM component history, after splitting at an elapsed time of 130 hours the number of components was nearly constant at eight components due to merging until 280 hours. After that, it reached nine components and was constant until the end of the simulation. For v_{EXP} , the response is similar except that a consistent nine components were present from 180 hours to 331 hours before a merge was performed near the end. The GMM prediction with v_{MSE} finished with seven components at the end of the propagation time, while the GMM prediction with v_{EXP} finished with nine components. The information lost for each merging operation is shown in Fig. 5(b) where the loss of information remained consistent and below J_{NISE} value of 2.0×10^{-3} . The near-vertical line present at 179 hours for v_{EXP} and 281 hours for v_{MSE} indicates merging occurred successively over the two time steps.

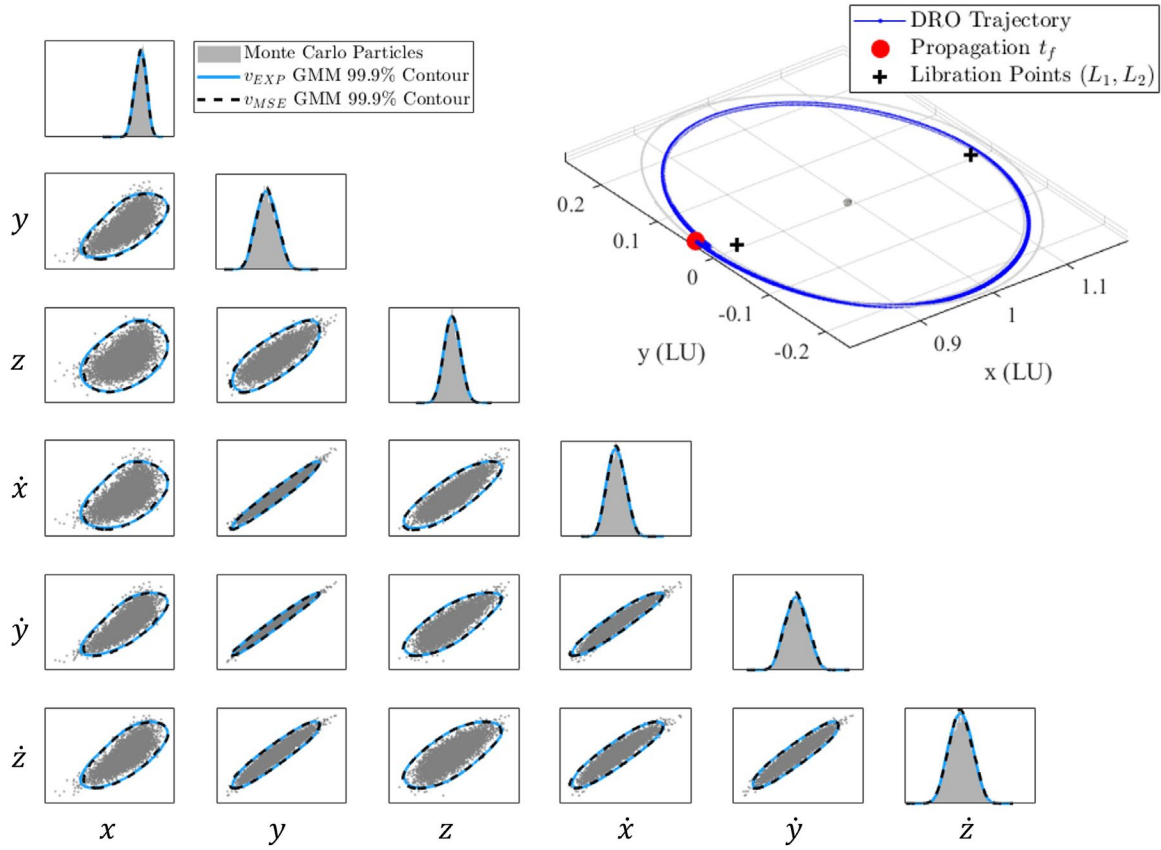


Fig. 4. DRO trajectory plot pairs that depict the projections of the propagated GMM PDF using the adaptive mixture algorithm with split criteria v_{MSE} and v_{EXP} , displayed at the 99.9% probability contour. The GMM PDF with split criterion v_{MSE} is shown by the black dashed line, while the v_{EXP} is shown with the blue line. The Monte Carlo particles are shown by the grey points.

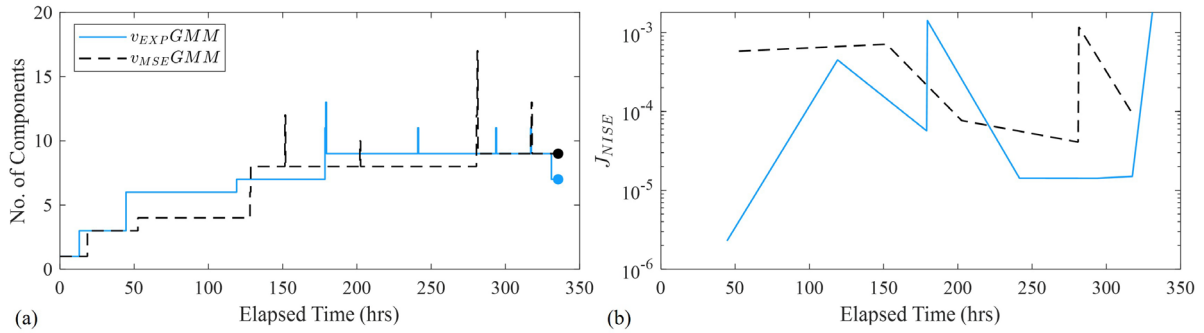


Fig. 5. (a) Plot of the number of components in the propagated GMM for the DRO trajectory using both v_{EXP} and v_{MSE} split criteria shown in blue and black dashed lines, respectively. (b) Plot of the J_{NISE} corresponding to each time step in which merging was performed for both v_{EXP} and v_{MSE} based splitting criteria.

A summary of the performance is displayed in Table 1, which presents the maximum J_{NISE} reached for all merging operations, the LAM at t_f , and NNA at t_f . The NNA at the prediction end time indicates that relatively few particles failed to associate. The LAM between the two GMMs is relatively similar, where the v_{EXP} -based GMM is somewhat more diffuse with lower NNA or number of particles rejected.

Table 1. Comparison of performance metrics for the DRO trajectory. The maximum J_{NISE} is shown for $t \in [t_0, t_f]$, while LAM and NNA are shown at t_f for predicted GMM for each v_{EXP} and v_{MSE} splitting criteria.

	$\max J_{NISE}$	LAM	NNA (%)
v_{EXP} GMM	1.796×10^{-3}	1.619×10^{18}	0.13
v_{MSE} GMM	1.176×10^{-3}	1.631×10^{18}	0.18

4.2 Lunar Transfer Orbit Trajectory

For the LTO, Fig. 6 displays the pair-wise projections of the v_{MSE} - and v_{EXP} -based GMMs. In comparison to the DRO case, the LTO distribution is vastly stretched, requiring a GMM with numerous components for an accurate representation. Similar to Fig. 4, the main diagonal in Fig. 6 presents the PDF over each dimension represented by both the v_{MSE} and v_{EXP} GMMs with the Monte Carlo distribution. It is seen from the main diagonal of plots that the mixture captures the majority of the uncertainty presented by the Monte Carlo particles for each dimension $\mathbf{x} \in \mathbb{R}^{n_d}$. Additional detail is highlighted in the red overlays for a few selected projections. Examples where the edges of the mixture do not capture the entirety of the Monte Carlo particles are shown for the $\dot{x}$ - y , $\dot{z}$ - y , and $\dot{x}$ - z projections. The overlay of the $\dot{y}$ - z projection is zoomed over the center of the distribution to show that the GMM encapsulates the Monte Carlo distribution tightly. There are some differences noted along the edges of the GMMs (i.e. the $\dot{x}$ - z and $\dot{z}$ - y projections) where the v_{MSE} GMM extends to a larger volume than the v_{EXP} GMM.

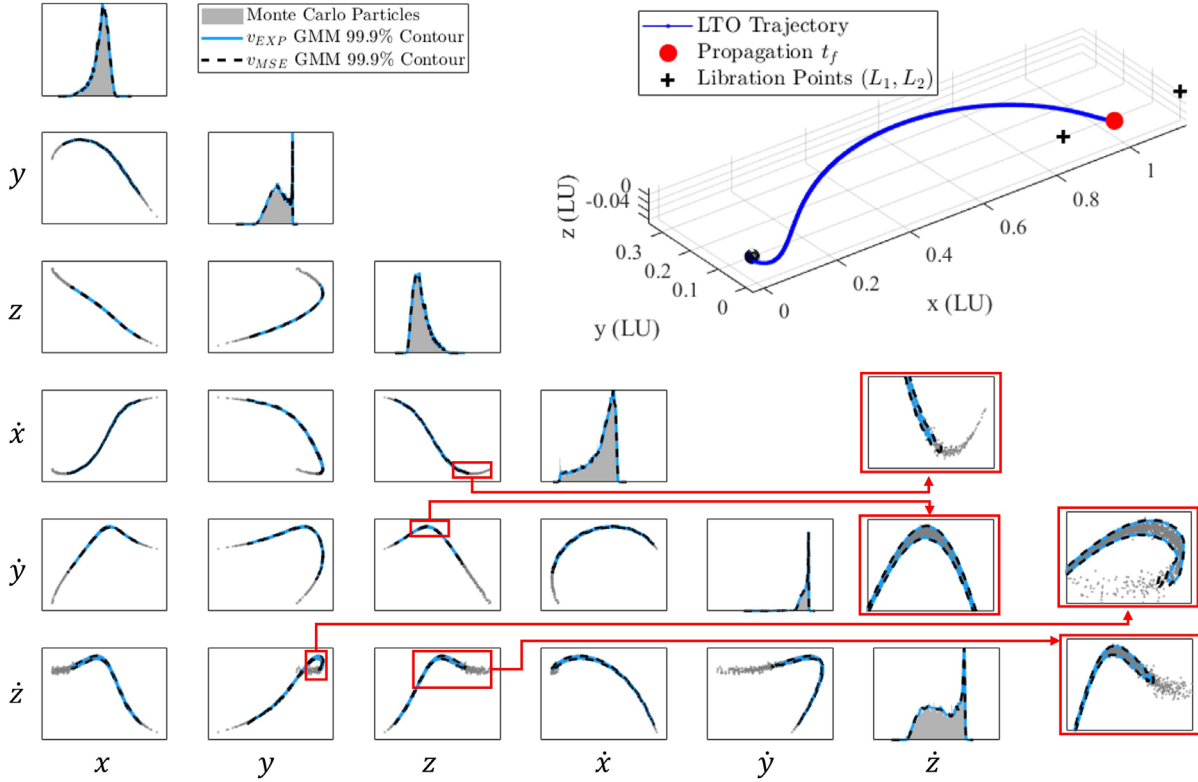


Fig. 6. LTO trajectory plot pairs that depict the projection of the propagated GMM using the adaptive mixture algorithm with split criteria v_{MSE} and v_{EXP} for the 99.9% probability contour. The main diagonal is the PDF for the single dimension for both Monte Carlo particles and the respective GMMs. The $\dot{x}$ - z , $\dot{y}$ - z , $\dot{z}$ - y , and $\dot{x}$ - z projections highlighted in red are further highlighted to show additional detail. The upper right trajectory plot depicts propagation end time t_f with the red point for which the respective GMMs are shown.

For the LTO trajectory, most of the splitting and merging behavior occurred near the Moon at the end of the trajectory. Fig. 7(a) illustrates the number of components for each mixture, where a zoomed section of the plot is included to show the structure near the end time. A large fluctuation in the number of components indicates sensitivity with respect to both statistical-linearization-based criteria for splitting. It is seen that the v_{MSE} GMM reaches a higher number of components before v_{EXP} GMM. For each merge executed, the information loss in terms of J_{NISE} is shown in Fig. 7(b),

where the region with higher merging frequency is highlighted and detailed. The maximum J_{NISE} reached for v_{MSE} based GMM is lower than the J_{NISE} reached for the v_{EXP} based GMM as captured in Table 2.

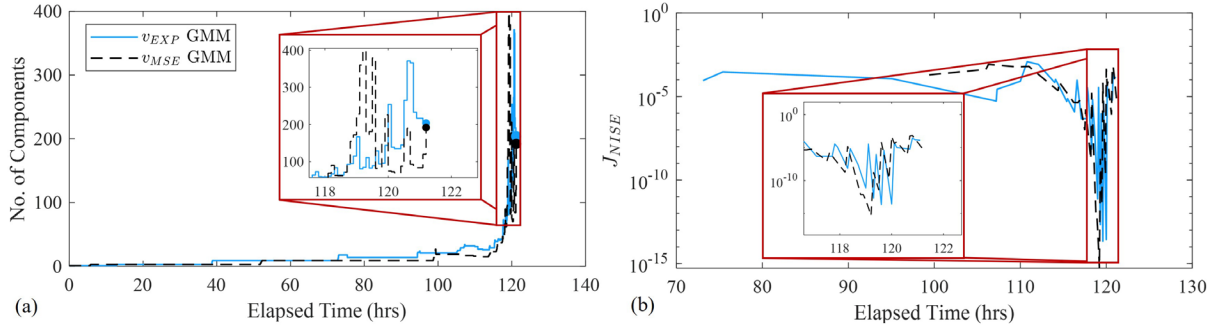


Fig. 7. (a) Plot of the number of components over the propagated GMM for the LTO trajectory using both v_{EXP} and v_{MSE} split criterion shown in blue and black dashed lines, respectively. (b) Plot of the J_{NISE} corresponding to each time step in which merging was performed for both v_{EXP} and v_{MSE} based splitting criteria.

Additional detail on the merging for each mixture is shown in Fig. 8, where, for each time step in which merging occurred, the J_{NISE} sensitivity was calculated for a D_{uKL} value up to 0.2. An overlay of the mixture J_{NISE} depicted in Fig. 7(b) is included in Fig. 8 by the magenta line. Fig. 8(a) depicts the J_{NISE} sensitivity from v_{MSE} based GMM and Fig. 8(b) illustrates a J_{NISE} surface from v_{EXP} based GMM. When the J_{NISE} surface is ‘peaky’, that means it is sensitive, such that a small increase in merging will likely cause excessive information loss to occur. In both cases, over the course of the propagation, an increase in the J_{NISE} sensitivity is seen, while the surface from the v_{EXP} -based GMM reaches a higher values than the v_{MSE} -based GMM. The merging depicted by the magenta lines in both Fig. 8(a) and Fig. 8(b) show a relatively smaller amount of information loss compared to rest of the surface. The merging performed during the simulation reduced the number of accumulated components at the cost of relatively low information loss, as indicated by the J_{NISE} time series when compared to each respective J_{NISE} surface. These plots indicate that if more aggressive merging is pursued, higher losses of information would be incurred, especially when using the v_{EXP} trigger.

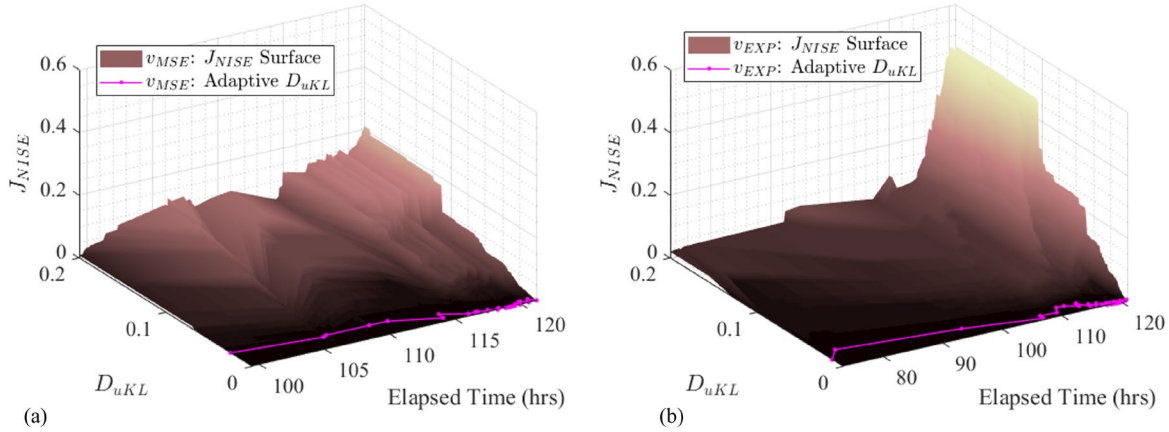


Fig. 8. Plot of the J_{NISE} surface over D_{uKL} and elapsed time for the LTO case in which (a) the v_{MSE} and (b) v_{EXP} were used for splitting criterion. The result of the adaptive merging is shown by the magenta dotted line.

A summary of the results is presented in Table 2, where in this case the LAM and NNA indicate a few differences between the splitting criteria. The v_{EXP} GMM captured an order of magnitude higher LAM while rejecting 0.18% more particles compared to the v_{MSE} GMM.

Table 2. Comparison of maximum J_{NISE} , LAM and NNA for the LTO trajectory at t_f for predicted GMM for each v_{EXP} and v_{MSE} splitting criteria.

	$\max J_{NISE}$	LAM	NNA (%)
v_{EXP} GMM	1.212×10^{-3}	7.316×10^{17}	1.38
v_{MSE} GMM	8.385×10^{-4}	4.491×10^{16}	1.20

4.3 Near Rectilinear Halo Orbit

In Fig. 9 the GMM projections are shown for the NRHO trajectory for both v_{MSE} and v_{EXP} GMMs. Comparing the main diagonal plots here to the LTO and DRO cases, the v_{MSE} and v_{EXP} GMMs are less precise, as seen when comparing the PDF for each dimension against the Monte Carlo particles. The PDF spatial projections also show that the resulting GMM is less accurate in overlapping the particle distribution. The v_{MSE} GMM is particularly conservative in that the volume of the GMM is much larger than needed to represent the distribution. Additional detail is shown for z - y , $\dot{x}$ - z , $\dot{z}$ - z , and $\dot{z}$ - $\dot{z}$ projections, where it is seen that the GMMs capture the majority of the Monte Carlo particles in the denser regions of the distribution.

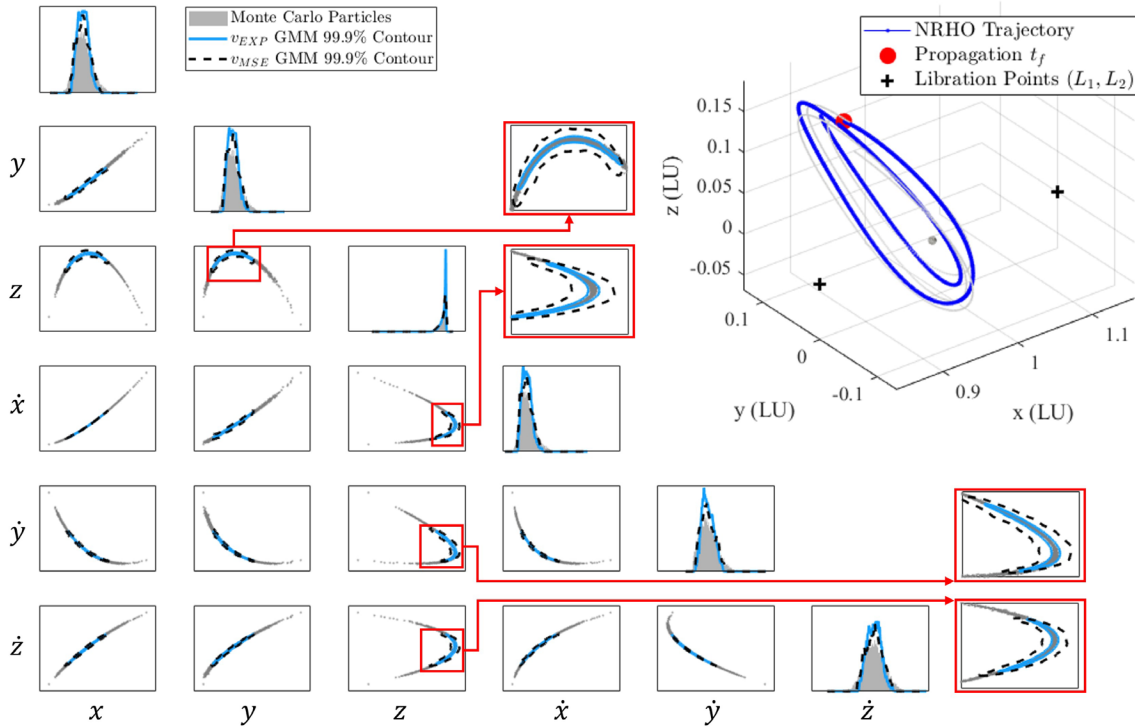


Fig. 9. NRHO trajectory plot pairs that depict the projection of the propagated GMM using the adaptive mixture algorithm with split criteria v_{MSE} and v_{EXP} for the 99.9% probability contour. The main diagonal is the PDF for the single dimension for both Monte Carlo particles and the respective GMMs. The z - y , $\dot{x}$ - z , $\dot{y}$ - z , and $\dot{z}$ - z projections highlighted in red are further inspected to show additional detail. The upper right trajectory plot denotes the propagation end time t_f , at which the respective GMMs are shown, with the red point.

In Fig. 10(a), the time history of the number of components for both GMMs is shown. This trajectory has two close approaches to the Moon. The first close approach to the Moon, near elapsed time of 63 hours, is revealed by the first net increase in number of components. During the time between first and second approach, the adaptive GM algorithm maintains a nearly net constant number of components. Due to a change in the strength of nonlinearities between the first apolune and second close approach to the Moon, the PDF requires a larger number of components for appropriate representation at the latter approach. The second close approach near elapsed time of 300 hours causes a consistent net increase in the number of components. Similar behavior in both splitting and merging is shown for v_{MSE} and v_{EXP} . The loss of information according to each merge is illustrated in Fig. 10(b), where a high frequency of merging is performed during the second close approach to the Moon for both GMMs.

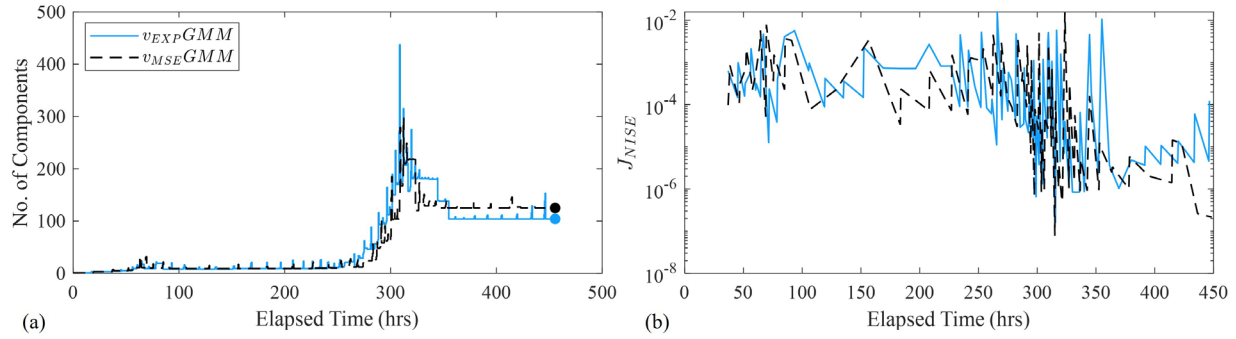


Fig. 10. (a) Plot of the number of components over the propagated GMM for the NRHO trajectory using both v_{EXP} and v_{MSE} split criterion shown in blue and black dashed lines, respectively. (b) Plot of the J_{NISE} corresponding to each time step in which merging was performed for both v_{EXP} and v_{MSE} based splitting criterion.

The J_{NISE} surfaces are presented for the NRHO case in Fig. 11. These J_{NISE} surfaces describe the information loss due to increased merging with D_{uKL} up to 0.2 for each time step in which merging occurred. Fig. 11(a) displays the surface for the v_{MSE} -based GMM and Fig. 11(b) is the surface for the v_{EXP} -based GMM. In both surfaces, an increase in the J_{NISE} sensitivity is seen past the second close approach to the Moon at approximately 300 hours. The adaptive D_{uKL} demonstrated by the magenta line in both Fig. 11(a) and Fig. 11(b) depicts the lines represented in Fig. 10(b) for both GMMs. The adaptive merging algorithm achieves a reasonable amount of merging to keep the number of components tractable, while avoiding a large loss of information.

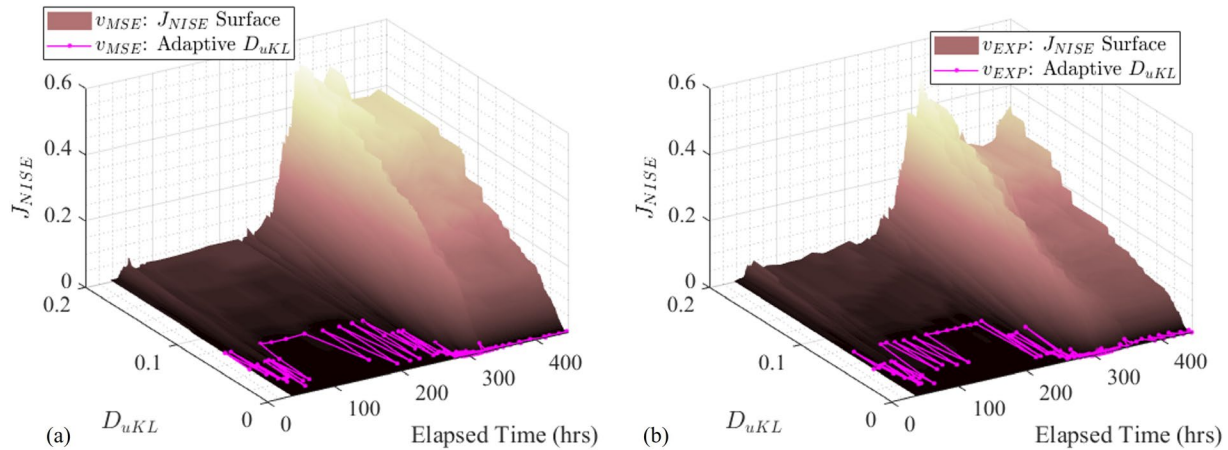


Fig. 11. Plot of the J_{NISE} surface over D_{uKL} and elapsed time for the NRHO case in which (a) the v_{MSE} and (b) v_{EXP} were used for splitting criteria. The result of the adaptive merging is shown by the magenta dotted line.

A visualization of the evolution of the GMM is presented in Fig. 12. The center plot shows the 3D trajectory with four points along the timeline identified as t_a through t_d where snapshots of the GMM are shown. Each snapshot displays the predicted v_{MSE} GMM in component form with the Monte Carlo particles in the background. For large portions of the trajectory the PDF represented by the Monte Carlo particles is approximately Gaussian. With the glaring exception being the lunar close approaches. The non-Gaussian shape shown at an elapsed time of 293 and 308 hours (t_c and t_d) is consistent with sensitivity shown in Fig. 11. Snapshot t_d highlights why a very large number of components is necessary to capture the true uncertainty distribution. Due to the interaction with lunar gravity, the uncertainty distribution is spread thinly over vast distance of $\sim 50,000$ km.

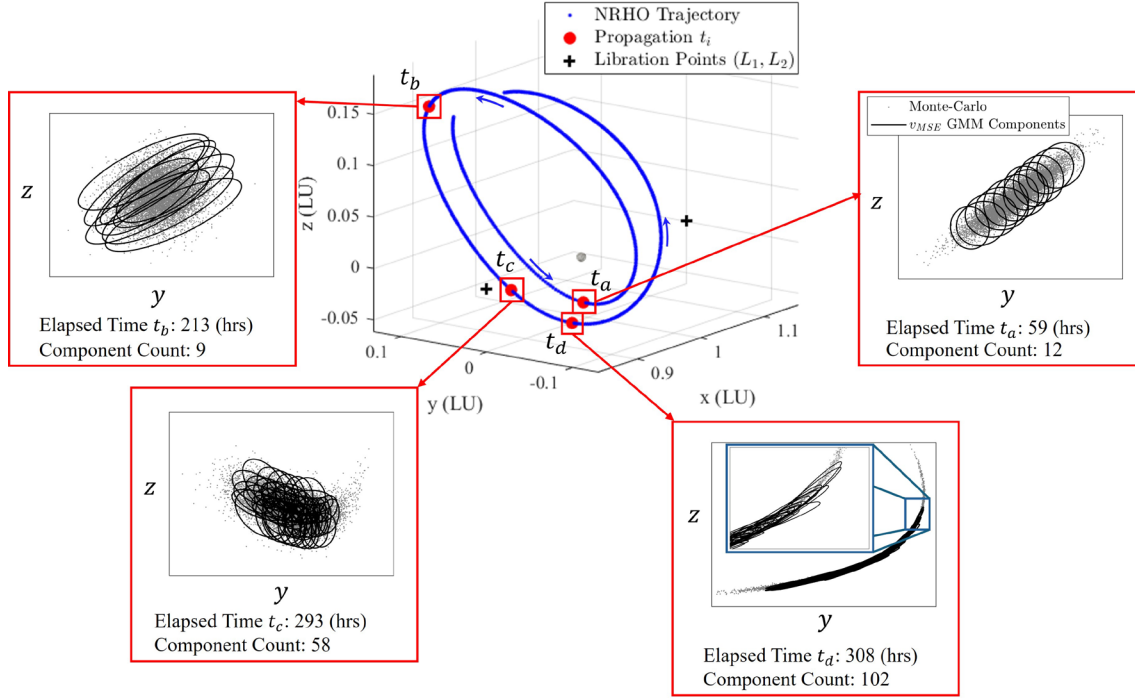


Fig. 12. Illustration of the evolution of the predicted GMM for the NRHO trajectory up to elapsed time of 293 hours (near second lunar close approach). The red points to various snapshots of elapsed times (t_a to t_d) for the NRHO trajectory. Each snapshot contains z-y projection plots of the predicted v_{MSE} GMM components for the 99.9% probability contour with the Monte Carlo points.

The summary in performance metrics for the NRHO case is shown in Table 3. The maximum J_{NISE} achieved by the v_{MSE} - and v_{EXP} -based GMMs is relatively similar. The v_{EXP} GMM achieved greater than one order of magnitude for LAM, while the number of particles not associated was higher. It is evident that the v_{MSE} -based GMM provided a more diffuse PDF, which is a similar result to that shown in the LTO case.

Table 3. Comparison of maximum J_{NISE} , LAM and NNA for the NRHO trajectory at t_f for predicted GMM for each v_{EXP} and v_{MSE} splitting criteria.

	$\max J_{NISE}$	LAM	NNA (%)
v_{EXP} GMM	1.578×10^{-2}	4.972×10^{16}	6.84
v_{MSE} GMM	1.537×10^{-2}	2.754×10^{15}	2.22

4.4 Results Summary

A summary of the performance metrics for the DRO, LTO and NRHO trajectories is shown in Table 4. For the same splitting and merging parameters, the LAM decreases over the set of trajectories and NNA increases. The LTO and NRHO trajectories displayed a higher (better) LAM for predicted GMM based on v_{EXP} . However, the v_{EXP} criteria produced a higher (worse) NNA value in the LTO and NRHO trajectories. In the projection plots produced in Fig. 9, the predicted GMM based on the v_{MSE} splitting metric had a larger volume over the projections shown. The LAM mirrors this observation by having a lower value, indicating a relatively more diffuse PDF represented by the GMM. Because the v_{MSE} GMM was more diffuse and spread out, it also associates more particles, leading to a lower NNA result.

A contributing factor in the difference between v_{MSE} and v_{EXP} is the sensitivity of each splitting metric. The v_{EXP} splitting metric tends to be more sensitive, which promotes splitting earlier and more often than the v_{MSE} splitting metric, as seen in Fig. 7(a) and Fig. 10(a). The increased splitting from v_{EXP} allows the GMM to capture higher resolution and leads to a more accurate GMM PDF compared to the GMM PDF from v_{MSE} splitting metric. The merging performance metric J_{NISE} has the largest value for the NRHO trajectory example from Table 4, because the

mixture is most prone to losing the largest amount of information due to merging during a close approach as depicted in Fig. 11 around the 300 hour mark.

Table 4. Summary of maximum J_{NISE} , LAM and NNA for the DRO, LTO and NRHO trajectories at t_f for predicted GMM for each v_{EXP} and v_{MSE} splitting criteria.

		$\max J_{NISE}$	LAM	NNA (%)
DRO				
	v_{EXP} GMM	1.796×10^{-3}	1.619×10^{18}	0.13
	v_{MSE} GMM	1.176×10^{-3}	1.631×10^{18}	0.18
LTO				
	v_{EXP} GMM	1.212×10^{-3}	7.316×10^{17}	1.38
	v_{MSE} GMM	8.385×10^{-4}	4.491×10^{16}	1.20
NRHO				
	v_{EXP} GMM	1.578×10^{-2}	4.972×10^{16}	6.84
	v_{MSE} GMM	1.537×10^{-2}	2.754×10^{15}	2.22

5 DISCUSSION

This paper proposed an adaptive Gaussian mixture model (GMM) algorithm, which incorporates a split-merge methodology. The splitting mechanism used is derived from the statistical linearization error, where two metrics are selected for testing. The bounded nature of the two metrics allows an operator to tune the splitting sensitivity easily. The adaptive GMM algorithm was used with a uniform set of GMM split-merge parameters for all three cases to demonstrate the performance envelope over a range of trajectories. For all three cases, the split-merge algorithm captured the majority of the Monte Carlo particles. Notably the difference in performance between v_{MSE} and v_{EXP} can be attributed to the sensitivity of each splitting metric. Choice of splitting criterion depends on the level of conservatism desired by the analyst. The merging analysis of the LTO and NRHO trajectories indicated that there are regions of a trajectory in which the GMM can be sensitive and prone to large loss of information from even a small amount of merging. The mixture size optimization routine provides a tunable lower bound on GMM component number that successfully maintains the information content for uncertainty propagation.

In the limit, a GMM can mimic any distribution if the cost to propagate the components is managed. The proposed approach adaptively controls the number of components needed to accurately represent the uncertainty in cislunar applications. This fidelity of uncertainty representation is significantly better than what would be possible with a standard EKF or UKF filter prediction. The method succeeded on trajectories with significantly differing dynamics, while requiring minimal tuning or specialization of the method. Essentially the same parameters were used to generate all results in this paper, making the method very attractive for operational settings in which knowledge of the orbit is not necessarily known *a priori*.

There are some caveats for the method that especially manifest themselves in highly nonlinear regions, such as those seen in the NRHO close approaches to the Moon. Generally, splitting early and often can alleviate accuracy issues at the cost of a high number of components. The following is a discussion of some of the issues that were encountered. A) Both the acts of splitting and merging, to various extents, discard the accumulated nonlinearities found in the sigma points of a current component. B) If splitting is delayed, the new components may be introduced too far away from the original, thus introducing error. In the extreme, consider a component that is stretching due to a close encounter with a planetary body, causing significant skewness (the well known “banana” shape in Cartesian coordinates). Due to the symmetric assumptions in a split, the skewness would be discarded and the new components could be introduced far from the actual trajectory they should be following. C) An additional attribute of splitting can be problematic in extreme cases. Most splitting libraries are homoscedastic; i.e., the new components all have the same variance. This assumption allows for the sigma points created from the new components to be more distant from the original mean than is appropriate relative to the original sigma points. Again, in the extreme, these new points are no longer close enough to be experiencing the same dynamics, which can introduce perturbations that are quickly magnified.

Future work on the topic will seek to refine the accuracy of the predicted GMM. Improvements can be made to several fundamental components of the algorithm. The splitting mechanism relies on the choice of a splitting direction, and there are cases that may require a more effective approach to capture sensitive dynamics. A merging operation can

often reset any accumulated nonlinearity for components in which merging was performed. Additional constraints could be implemented to prevent merging in some parts of the GMM that are sensitive to accumulated nonlinearity.

6 ACKNOWLEDGEMENTS

This work was funded by the Air Force Research Laboratory under contract FA9453-23-C-X002.

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